

Peer skill identification and social class: Evidence from a referral field experiment*

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Abstract

Cognitive and social skills are both increasingly valued in the labor market, but social skills are difficult to observe. In the absence of observable signals, peer assessments can be valuable screening tools. We study how well individuals identify productive peers across cognitive and social skills in a lab-in-the-field experiment with 849 university students. After students interact for an entire term, we collect incentivized skill measures from all classmates. We then ask for referrals of the most productive peers in each skill and incentivize referrals based on the nominee’s score in each skill measure. To examine potential social class barriers in referrals, we randomly assign half of the participants to receive additional incentives for identifying high-skilled peers from low-socioeconomic status. We find that peers can successfully identify cognitive skills but not social skills of their classmates. There is only evidence of a bias against low-SES peers in unique cogni-

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tive skill referrals, and the treatment incentives helps mitigate it. Our findings suggest that the accuracy of peer assessments varies substantially across skill dimensions and appropriate changes in the incentivization structure can make peer assessments robust to existing biases.

JEL Classification: C93, D03, D83, J24

Keywords: network homophily, labor markets, performance evaluation, hiring screening, human capital, incentive mechanisms, workplace diversity, academic performance, socioeconomic barriers, information asymmetry

1 Introduction

Evaluating the productivity of others is a standard feature of labor markets. Employers assess job candidates, managers evaluate workers for promotion, and team leaders select collaborators based on beliefs about others’ capacity to perform well in different tasks. Whenever observable productivity signals such as test scores or past experience are available, decision-makers rely on those to make accurate evaluations. But such signals are scarce for tasks that are interpersonal in their nature and difficult to quantify. In these settings, peer assessments akin to referrals can be a particularly strong screening tool which combines cost-efficiency and accuracy, as sustained interactions among people who work together provide opportunities to directly observe each other’s productive qualities in various domains.

However, identifying productive peers across a multitude of productivity dimensions is not straightforward. First, peers could accurately assess productivity in one dimension, but they may struggle to evaluate it in another because of its harder to observe nature. Cognitive and social (interpersonal) skills are two such dimensions of human capital that are increasingly rewarded in labor markets (Deming, 2017, 2023). Second, biases in productivity beliefs can lead to systematic deviations in assessment accuracy. The case for low-socioeconomic status (low-SES) individuals is particularly concerning, as peers may systematically underestimate their abilities due to stereotypes or lack of information.

Such biased assessments could contribute to their worse labor market outcomes despite having the necessary skills (Stansbury & Rodriguez, 2024).

The overall purpose of this paper is twofold: To evaluate how accurately peers identify productive others in cognitive and social skills, and whether disadvantaged low-SES individuals face barriers in selection when peers assess productivity across these skills.

We conducted a lab-in-the-field experiment in a Colombian university to answer these questions. After interacting for an entire term (about 4 months) in small classrooms (average 26 students per class), we collected incentivized cognitive and social skill measures from all participants to obtain objective productivity distributions. Participants then assessed classmates’ productivity across these dimensions by making referrals, allowing us to compare referred peers to those who were not. We incentivized referrals by bonuses contingent on the nominee’s score in the skill measures. Nominees did not receive any benefit from being referred. Both features allowed us to rule out concerns of potential social transfers (i.e., nepotism or favoritism) and reputational costs typical in the referral literature (see for example Bandiera, Barankay, and Rasul (2009); Witte (2021)). Once we abstracted away from these elements, the referral decision became one of measuring productivity beliefs through nominated candidates.

Even in an incentivized setting like ours, biases about low-SES individuals could be at play because of the underlying beliefs classmates hold about their productivity. To address this we designed two treatments. In the **Baseline** treatment, we gave pure performance incentives to referrals regardless of social class. Participants in the **Quota** treatment received additional incentives to identify high-skilled low-SES peers. To be able to make comparisons within the same referral choice sets, we assigned half of the participants within each classroom to either treatment. This setup allows us to assess how well incentives mitigate the said biases in peer productivity beliefs across the different referral behaviors that we observe.

Our first goal is understanding how well peers identify cognitive and social skills of their classmates under pure performance incentives at **Baseline**. We find that peers have distinct screening abilities for skills, and use different types of referral strategies

73 because of it. Specifically, peers successfully identify cognitive skill but not social skill 73
74 of their classmates. They also frequently refer the same peers for both skills, at rates 74
75 much higher than the actual overlap between those who are productive at both cogni- 75
76 tive and social skill. For this reason we separately analyzed the three referral types: 76
77 Those made in common for both skills, and those made uniquely for cognitive or social 77
78 skill. Common referrals for both skills identified classmates with higher grades but not 78
79 higher skills. This suggests an observable proxy such as academic performance influences 79
80 peer productivity assessments in the absence of credible skill information. For unique 80
81 cognitive skill referrals, both grades and measured cognitive skill are equally good pre- 81
82 dictors. Unique social skill referrals are not predicted by either academic performance 82
83 or social skill, suggesting that social skills might be less observable in classroom settings 83
84 or require different measures to evaluate accurately. These findings reveal a nuanced 84
85 picture of how peer assessments of productivity may depend on how discernible the skill 85
86 in question is, and how they can be influenced by the availability of other observable 86
87 proxies for productivity. 87

88 We find limited support for a bias affecting low-SES individuals. Of the three referral 88
89 types, we find bias only in unique cognitive skill referrals when accounting for peer 89
90 skills. This characterizes the decisions of about 75% of participants who made at least 90
91 one unique cognitive skill referral, and about half of all cognitive skill referrals overall. 91
92 The **Quota** treatment mitigates the bias for this subset of referrals, while not changing 92
93 the referral rates of low-SES individuals for the rest of the referral strategies that were 93
94 not biased in the first place. There is also no meaningful efficiency-equity tradeoff 94
95 affecting productivity of peers referred in the **Quota** treatment. Our findings show peer 95
96 productivity assessments are robust to salient differences between social classes, and 96
97 provide evidence that existing biases can be remedied with changes in the incentivization 97
98 structure without comprimising productivity. 98

99 Our paper contributes to various strands of the literature. First, we contribute the 99
100 literature on referral experiments that strives to understand how referrals help screen- 100
101 ing for productive workers. Past work provides causal evidence that peer productivity 101

assessments using referrals bring in productive workers (Pallais & Sands, 2016), and that performance-contingent incentives lead to improvements in the productivity of referred candidates (Beaman, Keleher, & Magruder, 2018; Beaman & Magruder, 2012). These studies allow referrals to be made from different candidate pools where referrers are free to nominate any candidate, and as a result confound screening ability with advantages arising from access to different candidate pools (Montgomery, 1991). We implement common choice sets for referrals which allow us to isolate peers’ true screening ability and enable straightforward comparison between experimental treatments in terms of referral choice sets. Our paper complements the literature on referral experiments by providing causal evidence that peers have skill-dependent screening abilities that go beyond the differences in candidate pools under performance-contingent incentives.

Second, we contribute to the growing body of work on the relevance of noncognitive skills in the labor market. This literature examines dimensions of human capital such as patience, self-control, conscientiousness, teamwork, and critical thinking that contribute positively to labor market returns (Heckman & Kautz, 2012; Heckman, Stixrud, & Urzua, 2006; Lindqvist & Vestman, 2011; Weinberger, 2014). Among these, interpersonal skills are exceptionally relevant for labor market gains in the last two decades as a complement to cognitive skill (Deming, 2017, 2023). Yet, hiring firms report difficulties in assessing social skills in candidates, and applicants are willing to pay substantial sums to convey social skill feedback to employers (Bassi & Nansamba, 2022). We contribute to this literature with our peer productivity assessments across two dimensions of skills, and show that peers can identify cognitive skill but struggle to assess social skills. Our results suggest that referrals may be ineffective for screening attributes that are less visible or harder to proxy through standard productivity measures in the assessment environment.

Finally, we contribute to the literature on diversity considerations in referrals. Homophily¹ in referrals drives correlations among social groups’ employment and wages

¹A well-documented empirical consistency in sociology where individuals form ties more often with others who are similar to themselves across observable characteristics (McPherson, Smith-Lovin, & Brashears, 2006; McPherson, Smith-Lovin, & Cook, 2001).

(Calvo-Armengol & Jackson, 2004; Calvo-Armengol & Jackson, 2007), as individuals are more often tied to others with comparable socioeconomic status (Chetty et al., 2022b). Limited interaction across social classes due to spatial segregation is shown to drive at least some of the differences (Chetty et al., 2022a). In this context, efficiency of diversity treatments in endogenous networks may be constrained by availability. To counter this, we consider a socially diverse university setting where we use exogenously imposed networks, and required participants to refer among classmates. Anticipating differences in referral outcomes for low-SES individuals even when networks across social classes overlap by design, we introduced quota-like incentives as a treatment arm to increase referrals to low-SES peers.² Our findings complement the literature on biases in referrals (Beugnot & Peterlé, 2020; Hederos, Sandberg, Kvissberg, & Polano, 2024) by first showing the existence of a social class bias and then providing the causal evidence for targeted incentives that effectively reduce the bias in our setting without compromising productivity.

The remainder of the paper is organized as follows. Section 2 begins with the background and setting in Colombia. In Section 3 we present the design of the experiment, including the skill assessment, referral and guessing tasks. In Section 4 we describe the data and procedures. Section 5 discusses the results of the experiment. Section 6 concludes. The Appendix presents additional tables and figures as well as the experiment instructions.

2 Background and Setting

Our study takes place at UNAB, a medium-sized private university in Bucaramanga, Colombia with approximately 6,000 enrolled students. The university’s student body is remarkably diverse with slightly more than half of the students classified as low-SES. This diversity provides a unique research setting, as Colombian society is highly unequal

²We design the treatment incentives in inspiration from the success of gender quotas in the affirmative action literature (e.g., Balafoutas and Sutter (2012); Bertrand, Black, Jensen, and Lleras-Muney (2019); Niederle, Segal, and Vesterlund (2013)).

and generally characterized by limited interaction between social classes, with different socioeconomic groups separated by education and geographic residence.³ Despite significant financial barriers, many lower middle-class families prioritize university education for their children (Hudson & Library of Congress, 2010, p. 103), with UNAB representing one of the few environments where sustained inter-class contact occurs naturally.

In 1994, Colombia introduced a nationwide classification system dividing the population into 6 strata based on housing characteristics and neighborhood amenities.⁴ We use this exogenous cutoff as the measure of social class in our experiment: Students in strata 1 to 3 are categorized as low-SES, and those in strata 4 to 6 as high-SES (see Appendix Figure A.1 for a detailed stratum distribution of our sample).

We invite all students enrolled in two compulsory courses to participate in our experiment. Throughout the term, students meet weekly for three-hour sessions where attendance is mandatory. Both courses are university-wide graduation requirements which result in large variations in academic programs (see Appendix Table A.3) and socioeconomic backgrounds across the classrooms. This setup provides a unique opportunity for collaborative inter-class contact on equal status, whose positive effects on reducing discrimination are casually documented (Lowe, 2021; Mousa, 2020; Rao, 2019).

3 Design

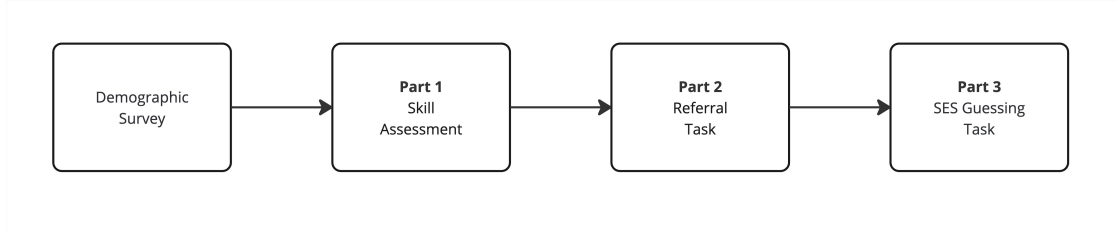
We designed an experiment to assess the peer screening ability for different skills and to measure biases related to social class. The study design consists of a single experiment

³Colombia has consistently ranked as one of the most unequal countries in Latin America (World Bank, 2024), with the richest decile earning 50 times more than the poorest decile (United Nations, 2023). This economic disparity is reflected by a highly stratified society with significant class inequalities and limited class mobility (Angulo, Gaviria, Páez, & Azevedo, 2012; García, Rodríguez, Sánchez, & Bedoya, 2015).

⁴Initially designed for utility subsidies from higher strata (5 and 6) to support lower strata (1 to 3), it now extends to university fees and social program eligibility. Stratum 4 neither receives subsidies nor pays extra taxes. This stratification system largely aligns with and potentially reinforces existing social class divisions (Guevara S & Shields, 2019; Uribe-Mallarino, 2008).

with sessions organized at the classroom level (see Figure 1). The instructions are provided in Appendix B.

Figure 1: Experiment Timeline



Note: Participants first complete incentivized skill tests, then refer classmates for skills. In the final part, they guess the social class of their peers. This order is implemented in all sessions.

3.1 Skill Assessment

To understand the basis for referral decisions, we collect objective measures of cognitive and social skills. These two distinct skills are crucial for the labor market and suitable to assess given classmates interact through the term. By measuring skills before the referral stage, we eliminated the need for referred students to take additional action. Participants perform two incentivized skill tests. They have 5 minutes to complete each test. We provide test-specific instructions and an example item before participants begin. Correctly solved items increase chances to earn a fixed bonus.⁵

We use Raven’s Progressive Matrices to measure cognitive skills (Raven, 1936; Raven, Raven, & Court, 1976). Raven’s test is a well-established measure of fluid intelligence, i.e., an individual’s capacity to reason and solve problems in novel situations independent of past knowledge (Schilbach, Schofield, & Mullainathan, 2016). In this test, participants see series of images where there is a pattern with a piece that has been intentionally removed. They are tasked with choosing the piece that completes the pattern among

⁵The tests are presented in a randomized order. No performance feedback is provided. Participants see one item at a time and cannot return to previous screens once they start a test. They are not required to answer items and can skip them if they choose to do so. We elicit beliefs about performance after each test.

available options. For each image, there is only one correct answer. We implement an 18-item version featuring increasingly difficult questions, with 6 response options for the first 9 items and 8 thereafter.

We measure social skills with the Multiracial Reading the Mind in the Eyes Test (MRMET) from Kim et al. (2022).⁶ The test is an established measure for the ability to recognize emotions in others, and it has been previously used in economic experiments (van Leeuwen et al., 2018; Weidmann & Deming, 2021; Zárate, 2023). MRMET is highly correlated with fluid intelligence as measured by Raven’s (Alan & Kubilay, 2025). It consists of photos of human faces portraying different emotions, cropped so that only the eye region is visible. Participants must choose the emotion that best describes the photo from the available answers. For each photo, there is only one correct answer and 4 response options. We administer the first 36 items in MRMET.

3.2 Referral Task

After the skill assessment, we create the referral task to screen for high skilled peers. For each skill, participants make incentivized referrals by nominating classmates. We first explain the measured skill accompanied by an example test item. On the following web page, we provide an alphabetically ordered list of all classmates. Participants make three referral choices per skill. They are instructed to exclude themselves from referrals. A classmate may be nominated once per triad. The order in which participants refer for a skill test is randomized. We incentivize referrals with classroom-level performance rankings. The three highest-scoring classmates are designated as the top 3 for a skill. Referrers are eligible for a fixed bonus for referrals among the top 3.⁷

We have two between subject treatments that varies the top 3 selection. In the **Baseline** treatment, the top 3 selection is based solely on performance ranking, regardless

⁶We choose MRMET because it is a race- and gender-inclusive test suitable for application in non-WEIRD (Western, Educated, Industrial, Rich, Democratic) populations like the one we sample from. The test is based on the original RMET (Baron-Cohen, Wheelwright, Hill, Raste, & Plumb, 2001).

⁷We solve ties among the top 3 randomly. We describe only the top 3 selection mechanism and provide no feedback about the top 3 composition to participants.

of other participant characteristics. The **Quota** treatment modifies the top 3 selection to prioritize low-SES individuals. We reserve the first spot in the top 3 for the highest-scoring low-SES peer, and assign the remaining two places based on performance (see Table 1). This guarantees at least one low-SES participant in the top 3 per skill. Participants are informed about the top 3 selection mechanism before making referral choices (Appendix Figure B.1 provides illustrations explaining the treatments). Assignment to the treatment is at the individual level within each classroom. This allows comparing the effect of the treatment while keeping the referral choice set constant.

Table 1: Places in the Top 3 according to composition rule

	Baseline	Quota
Merit-only	3	2
Reserved for low-SES	0	1

3.3 Socioeconomic Status Guessing Task

Participants make guesses about the anticipated SES of their classmates. We inform participants that a computer algorithm randomly selects three students belonging to strata 1, 2, or 3. They are tasked with nominating the people they believe the computer could choose at random (Appendix Figure B.2 provides the illustration explaining the task). Participants select three classmates from an alphabetically ordered list containing all their classmates. This task measures the ability to distinguish SES independent of test performance, as SES identification is relevant to our study.

4 Sample, Incentives, and Procedure

We invited 849 UNAB undergraduate students to participate in the experiment. Our final sample consists of 702 individuals who completed the study, resulting in an 83%

232 participation rate.⁸ We block randomized participants into treatments balancing gen- 232
233 der and social class. Table 2 presents key demographic characteristics and academic 233
234 performance indicators across treatments (Appendix Table A.1 compares the sample 234
235 and missing students). The sample is well-balanced between the **Baseline** and **Quota** 235
236 conditions and we observe no statistically significant differences in any of the reported 236
237 variables (all p values > 0.1). The sample is characterized by a majority of low-SES stu- 237
238 dents. About one-third of the sample are first-generation college students. The gender 238
239 distribution is balanced. The mean GPA of 3.95 is consistent across both treatments. 239

⁸The missing students did not come to class on the day of the experiment.

Table 2: Balance between treatment conditions

	Baseline	Quota	p
Low-SES	59%	55%	0.297
Female	52%	47%	0.195
Cognitive score (Raven's)	10.04	10.27	0.322
Social score (MRMET)	18.45	18.50	0.886
GPA	3.95	3.95	0.828
Entry exam score	61.85	62.17	0.638
Age	19.33	19.02	0.228
First generation	34%	37%	0.386
Ethnic minority	1%	3%	0.133
Rural community	30%	27%	0.308
Scholarship	1%	1%	0.916
# semesters at UNAB	3.18	3.17	0.916
N	368	334	702

Note: Low-SES indicates strata 1, 2, or 3. Cognitive score measures Raven's performance out of 18 questions. Social score reflects MRMET performance out of 36 questions. GPA indicates average grades out of 5. Entry exam represents the average score across reading, math, social sciences, and science components of Colombia's standardized university entrance exam ICFES. First generation indicates neither parent attended university. Rural community denotes residence in a non-urban area. p -values for binary outcomes are from two-sample tests of proportions; for continuous variables, from two-sample t -tests with equal variances. All reported p -values are two-tailed.

240 Participants could earn bonuses worth 100,000 Pesos (about 26 US Dollars) in each 240
241 part of the experiment. In the first part, we incentivized performance in the skill tests. 241
242 20% of participants were eligible for the bonus. We randomly picked one skill test for 242
243 each eligible participant and drew a number between 1 and 100. The participant received 243
244 the bonus if the percentage of correct answers in the selected test exceeded the drawn 244

number. Chances of earning the bonus increased with each correctly solved question by 5.5% (=1/18) for the Cognitive Skill test and by 2.78% (=1/36) for the Social Skill test.

In the second part, we incentivized referrals among the top 3 performers. 40% of participants were eligible for the bonus. We randomly selected one skill test and one referral for each eligible participant. The participant received the bonus if their referral was among the top 3. In the third part, we incentivized the correct identification of low-SES peers. 20% of participants in each classroom were eligible for the bonus. We randomly selected one guess for each eligible participant. The participant received the bonus if their guess correctly identified a low-SES peer. Draws for the bonuses were independent meaning participants could earn multiple bonuses.

Data collection occurred during the last two weeks of April 2024. Our local partner at UNAB, coordinated scheduled classroom visits and recruited research assistants to administer the experiment. Students present in class on the scheduled visit dates participated. Each classroom visit constituted a separate session. There were in total 35 sessions.⁹ Participants accessed the Qualtrics-based experiment using their smartphones during these visits. The median time to complete the survey was 20 minutes, with a compensation of \$26 for 117 lottery winners.

5 Results

5.1 Can peers screen cognitive and social skills?

Our first goal is understanding whether higher skilled individuals get more referrals. Because every referrer nominates 3 classmates per skill, analyzing only the extensive margin, i.e., whether an individual gets a referral, is not very informative.¹⁰ We consider the percentage share of referrals from individuals in **Baseline** condition as our dependent variable. This approach combines the intensive and extensive margins and also makes

⁹See Appendix Figures A.2a, A.2b and A.2c for the distribution of skills and GPA across classrooms and Appendix Table A.3 for diversity in program choices.

¹⁰Only 86 of the 849 students (10%) never get a referral for either skill.

comparisons across classrooms with different sizes easier.¹¹

Formally, we define the percentage share of referrals received by individual i from participants j in classroom c and in **Baseline** condition ($\forall j \in B_c$) for skill s as:

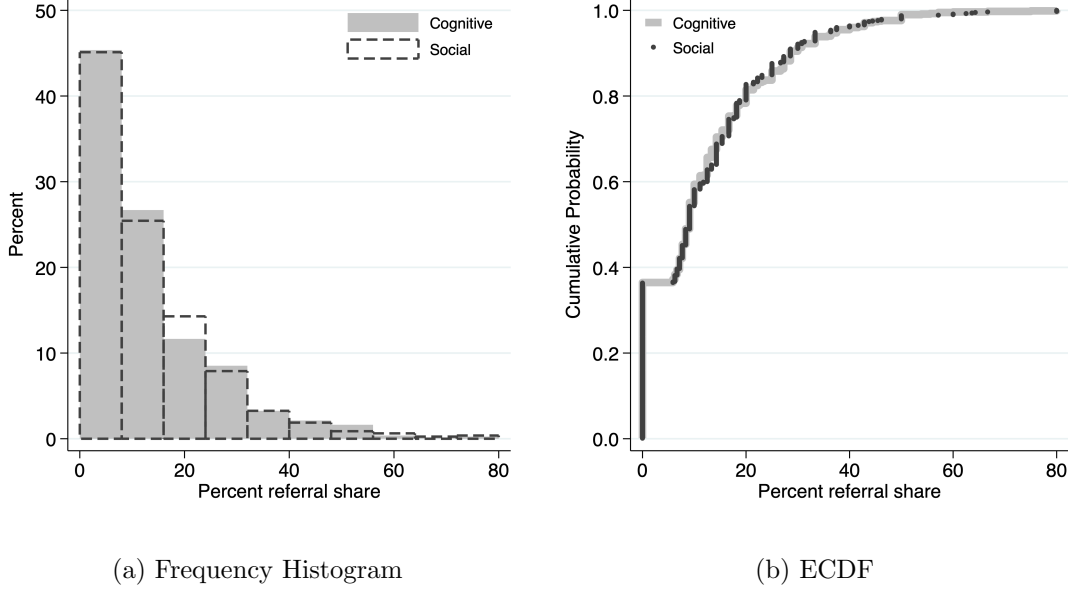
$$y_{ic}^s = \frac{\sum_{j \neq i} r_{ijc}^s}{n_c - \mathbb{1}(i \in B_c)} \times 100 \quad (1)$$

where n_c represents the number of participants in the **Baseline** condition in classroom c . The indicator r_{ijc}^s takes value 1 if participant j in the **Baseline** condition refers individual i for skill s , and 0 otherwise, and require both i and j to be in the same classroom c . The denominator $n_c - \mathbb{1}(i \in B_c)$ accounts for the maximum possible referrals that individual i could receive. If i is in the **Baseline** condition ($\mathbb{1}(i \in B_c) = 1$), we subtract one from n_c to account for the self-referral restriction.¹² This normalized measure represents the percentage of potential referrals actually received by each individual, adjusting for classroom size and treatment status. By construction, $y_i^s \in [0, 100]$ for all c , and we can compare referrals across classrooms of different sizes. Figures 2a and 2b present the distribution of our dependent variable.

¹¹The number of participants in a classroom mechanically drives the number of total referrals that could be received by an individual. By normalizing referrals we focus on differences within classrooms.

¹²33.8 percent of participants in the sample for cognitive and social skills self-referred, while explicitly instructed not to do so. In Appendix Table A.4 we compare the outcomes of those who self-refer. Self-referrers are more likely to be low-SES, and have significantly lower cognitive skill (0.2 SD) and GPA (0.25 SD). We rule out the hypothesis that self-referrers nominate themselves strategically. As self-referrers are not informative and add noise to our estimates, we drop these instances from our paired referral-referrer sample in subsequent analyses. Self-referring participants' remaining referral choices are kept in the dataset.

Figure 2: Distribution of referrals by skill in Baseline



Note: Figures show the percentage of referrals recieved from participants in the **Baseline** condition for cognitive and social skills. The left panel shows the frequency histogram and the right panel shows the empirical cumulative distribution function (ECDF). A two-sample Kolmogorov-Smirnov test shows no statistically significant difference between the share of referrals received across the skill distributions ($D = 0.0363$, $p = 0.668$).

Under performance pay in the **Baseline** condition, classmates with higher scores in the skill tests should collect more referrals if classmates can screen skills. Our independent variables are the standardized skill test scores. We estimate referral percentage shares y_i^s :

$$y_i^s = \alpha^s + \beta_1^s \text{Score}_i^s + \epsilon_i^s \quad (2)$$

Table 3 illustrates our first findings. Our preferred specification includes classroom fixed effects. The comparison of interest is the point estimates for different test scores. In column (2), a one standard deviation increase in cognitive skill score causes a 1.5 percentage point increase in the share of referrals received. On a base rate of 13%, this

is a modest increase of 11.5 percent. In column (4), 95% confidence intervals rule out that a one standard deviation increase in the social skill score results in more than a 0.8 percentage point difference in the share of referrals received.

Result 1 *Participants have difficulties screening skills in the **Baseline** condition, with modest screening ability for cognitive and no screening ability at all of social skill test scores.*

Table 3: Share of referrals received conditional on skill test score

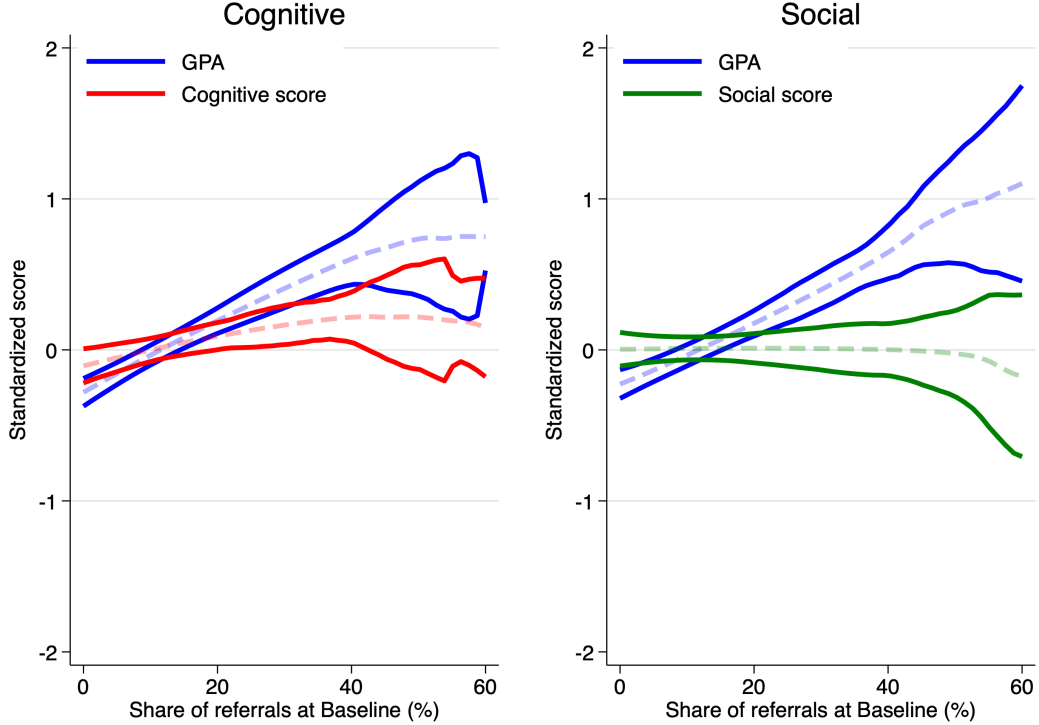
	Cognitive		Social	
	(1)	(2)	(3)	(4)
Score	1.197** (0.479)	1.497*** (0.464)	0.037 (0.474)	-0.080 (0.461)
Dep. var. mean	12.986	12.981	13.049	13.050
Classroom FE	No	Yes	No	Yes
R ²	0.008	0.116	0.000	0.100
Observations	665	665	665	665

Note: Classroom-level clustered standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Dependent variables are the percentage of referrals received relative to all referrals. “Score” refers to standardized test scores for cognitive and social skills. Sample restricted to 665 individuals for whom we have complete administrative and experimental data.

5.2 Grades as a proxy for skills

Absence of a clean skill-signal or the lacking the screening ability for skills may have pushed participants to refer classmates using proxies of skills. We define proxies as peer beliefs about strong correlates for skills. A potential proxy for cognitive skill (i.e., “smart students”) would be the “students with good grades” in the classroom, as measured by GPA. Figure 3 illustrates the relationship between grades, skill score, and the share of referrals received.

Figure 3: Referral shares by GPA and skill test scores



Note: The left panel shows how GPA and cognitive skill scores vary with the share of cognitive skill referrals received, while the right panel shows the same for GPA and social skill score for the share of social skill referrals received. Solid lines indicate 95% confidence intervals and dashed lines indicate the means. Output is truncated at 60 percent of referral share for the sake of having meaningful confidence intervals.

303 The idea that grades signal cognitive skill is a common belief among researchers 303
304 and practitioners alike. Yet, cognitive skill and grades are far from perfectly correlated 304
305 (Heckman & Kautz, 2012; Heckman et al., 2006), and screening with such beliefs may not 305
306 lead to good referrals. Indeed, GPA correlates very weakly with skill test scores in our 306
307 sample (see Appendix Table A.2). We capture the screening behavior using proxies by 307
308 including the standardized GPA of referrals as an independent variable. We reestimate 308
309 referral percentage shares for the **Baseline** condition: 309

$$y_i^s = \alpha^s + \beta_1^s Skill_i^s + \beta_2^s GPA_i^s + \epsilon_i^s \quad (3)$$

Table 4 illustrates our findings. Our preferred specification includes classroom fixed effects. The comparison of interest is the difference between point estimates for skill test scores and GPA. In column (2), a one standard deviation increase in cognitive skill score causes a 1.1 percentage point increase in the share of referrals received when controlling for GPA. On a base rate of 12.8%, this is a comparable increase in magnitude of about 8.6 percent to our previous estimate in Table 3, and suggests cognitive skills have an independent effect on referrals. However, a one standard deviation increase in GPA causes a substantial 4.4 percentage point increase in the share of referrals received when controlling for cognitive skill score. This is an increase of four times in terms of magnitude (34 percent) when compared to cognitive skill, and suggestive of the extent to which academic performance is easier to screen among peers in our setting.

In column (4), 95% confidence intervals rule out that a one standard deviation increase in the social skill score results in more than a 0.3 percentage point difference in the share of referrals received. This is consistent with our previous estimate confirming participants cannot screen social skill scores. On the other hand, a one standard deviation increase in GPA causes a substantial 3.8 percentage point increase in the share of referrals received when controlling for social skill. This is a 30 percent increase in the share of referrals when including controls for social skill.

Result 2 *For both skills, we find strong evidence that grades act as a proxy for referral decisions.*

Table 4: Share of referrals received conditional on skill test score and academic performance

	Cognitive		Social	
	(1)	(2)	(3)	(4)
Score	0.873*	1.080**	-0.278	-0.527
	(0.467)	(0.455)	(0.460)	(0.409)
GPA	3.949***	4.364***	3.429***	3.789***
	(0.664)	(0.684)	(0.581)	(0.651)
Dep. var. mean	12.806	12.783	12.891	12.876
Classroom FE	No	Yes	No	Yes
R ²	0.095	0.204	0.064	0.165
Observations	665	665	665	665

Note: Classroom-level clustered standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Dependent variables are the percentage of referrals received relative to all referrals. “Score” refers to standardized test scores for cognitive and social skills. GPA is standardized to mean zero and unit variance. Sample restricted to 665 individuals for whom we have complete administrative and experimental data.

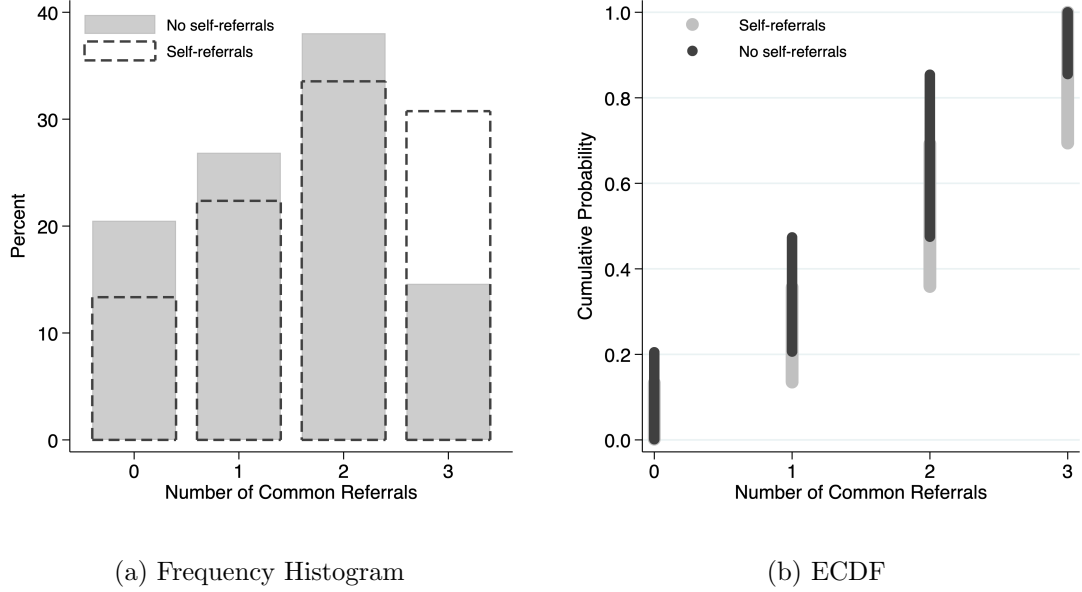
5.3 Types of Referrals

In this section, we expand on the diversity in referral choices to differentiate between referrers using GPA proxy and others. Despite having the opportunity to nominate up to six different classmates across two skills, referral choices are highly concentrated. Only 20% of participants made fully unique referrals, while the median participant nominated two classmates in common, effectively using four of their six referral slots for the same individuals. The concentration of referrals becomes even more pronounced when considering self-referrals which illustrate participants’ actual choices.¹³ 30% consistently chose themselves twice and the same two classmates across skills. Most participants

¹³Self-referrals were not valid and are excluded from the remaining analyses.

339 nominated two classmates in common for both skills, and picked themselves or someone 339
 340 else with almost equal probability. We visualize referral choices by plotting the number 340
 341 of common referrals made across skills in Figures 4a and 4b.¹⁴ 341

Figure 4: Common referrals between skills at Baseline



Note: Figures show the distribution of common referrals with and without self-referrals. The first bar (value of 0) indicates the share of participants with 6 unique referrals. The last bar (value of 3) indicates the share of participants with 3 identical referral choices across both skills.

342 With such a large share common referrals across skills, it is possible that participants 342
 343 believed classmates with a higher score in one skill would also have a higher score in the 343
 344 other. Would such beliefs be accurate? There is only modest ($\rho = 0.267$) correlation 344
 345 between the two skill test scores (see Appendix Table A.2). To understand whether 345
 346 making common referrals is strategic, we turn to the incentives. Participants were in- 346

¹⁴In Appendix Table A.5 we compare the characteristics of referrers who make unique referrals to those who made at least one common referral. Results suggest minimal differences in GPA, skills, and social class.

centivized to pick the top 3 performers for each skill to earn a fixed bonus. Looking at the characteristics of top skilled participants in Appendix Table A.6, we find that conditional on being among the top 3 for one of the skills, only 1 in 3 participants were in the top 3 for both skills. This suggests making more than 1 common referral across skills would decrease the chances to win the bonus.

A competing explanation for the amount of common referral choices between skills coupled with the notable difficulties in screening for skills would be that individuals who refer classmates twice for both skills are worse at screening. This implies the underlying heterogeneity in skill identification results in differential referral strategies where participants with a good signal for a skill choose to refer classmates only once for that skill, and those without a good signal use the grades proxy and refer classmates for both skills. We can test both hypotheses in our data: If “common” referrers -defined as those who refer an individual for both skills- are better at screening at least one of the skills, this gives more credence to beliefs about correlated skills. On the other hand, if “common” referrers are worse in skill identification compared to unique-referrers and use GPA proxy for referrals, we can infer that they have no additional information about skills.

We compare the outcomes of participants who receive common referrals from their classmates to those who receive unique referrals per skill. Formally, let indicator r_{ijc}^{common} take value 1 if individual j referred individual i for both skills. The percentage share of referrals received by individual i from participants in classroom c and in **Baseline** condition ($\forall j \in B_c$) is:

$$y_{ic}^{common} = \frac{\sum_{j \neq i} r_{ijc}^{common}}{n_c - \mathbb{1}(i \in B_c)} \times 100 \quad (4)$$

where n_c represents the number of participants in the **Baseline** condition in classroom c . The indicator r_{ijc}^{common} takes value 0 if participant j in the **Baseline** condition does not refer individual i for both skills. The denominator $n_c - \mathbb{1}(i \in B_c)$ accounts for the maximum possible “common” referrals that individual i could receive as before. Similarly, let $r_{ijc}^{s,unique}$ take value 1 if individual j referred individual i only for skill s . The percentage share of “unique” referrals received by individual i from participants in

classroom c and in **Baseline** condition ($\forall j \in B_c$) for skill s is:

$$y_{ic}^{s,unique} = \frac{\sum_{j \neq i} r_{ijc}^{s,unique}}{n_c - \mathbb{1}(i \in B_c)} \times 100 \quad (5)$$

and it follows that for any s , percentage share of “unique” and “common” referrals received by individual i from participants in classroom c and in **Baseline** condition ($\forall j \in B_c$) must add up to the total share of referrals received:

$$y_{ic}^s = y_{ic}^{s,unique} + y_{ic}^{common} \quad (6)$$

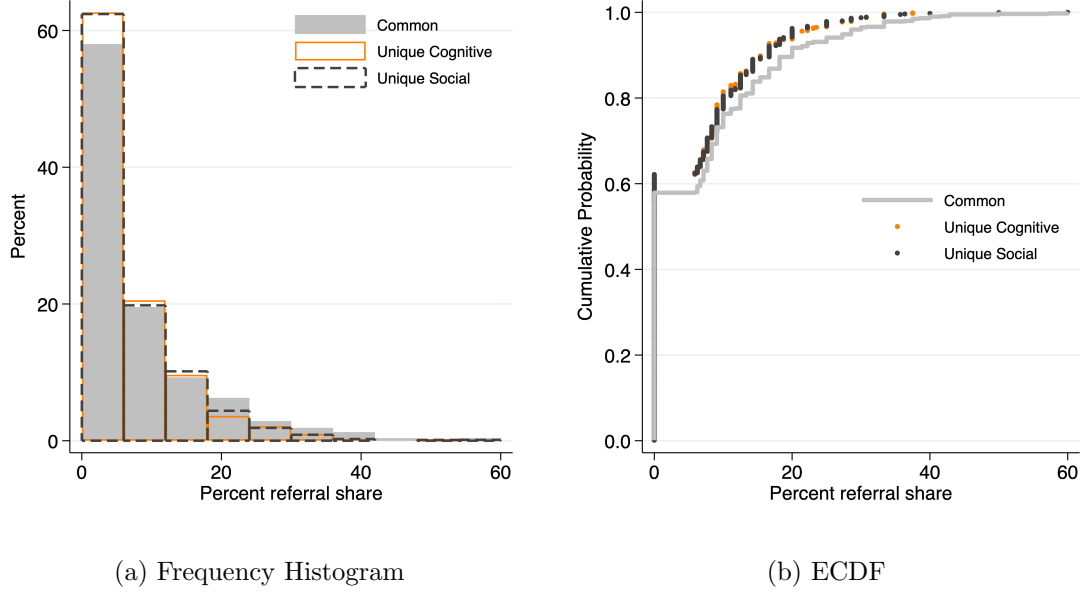
Table 5 shows the distribution of the referrals types in our sample. 37% of referrals fall under the type y_{ic}^{common} as pairs. This is equivalent to saying 54% of cognitive and social skill referrals were made in common. Figures 5a and 5b present the distributions of the three referral types.

Table 5: Distribution of Referral Types

	Frequency	Share (%)
Common	945	37.06%
Unique Cognitive	794	31.14%
Unique Social	811	31.80%
Total	2,550	100.00%

Note: Common referrals indicate the pair when the same classmate was referred for both cognitive and social skills. Unique referrals indicate when a classmate was referred for only one of the skills.

Figure 5: Distribution of common and unique referrals in Baseline



Note: Figures show the percentage of referrals recieved from participants in the **Baseline** condition depending on the referral type. The left panel shows the frequency histogram and the right panel shows the empirical cumulative distribution function (ECDF). Two-sample Kolmogorov-Smirnov tests show no statistically significant differences between the share of referrals received between “unique” cognitive and “unique” social referral distributions ($D = 0.0125$, $p = 1.000$) as well as “common” referrals ($D = 0.0602$, $p = 0.111$ for cognitive and $D = 0.0551$, $p = 0.177$ for social).

382 We regress Equation 3 for our three new dependent variables and report our findings 382
383 in Table 6. Our preferred specification includes classroom fixed effects. The comparison 383
384 of interest is the skill test scores and GPA estimates across columns. In column (2), we 384
385 find that a one standard deviation increase in GPA causes a 3.7 percentage point increase 385
386 in the share of “common” referrals received when controlling for skill test scores. This 386
387 is a substantial 50 percent increase on a base rate of 7.4%. Cognitive skills remain sta- 387
388 tistically insignificant and social skills show a marginally significant negative coefficient, 388
389 suggesting that participants who nominate the same individuals for both skills primarily 389

390 make referrals based on academic performance. 390

391 For participants who receive unique cognitive skill referrals, in column (4), we find 391
392 that a one standard deviation increase in GPA causes a 0.75 percentage point increase 392
393 in the share of referrals when controlling for cognitive skill test score. A one standard 393
394 deviation increase in cognitive skill test score causes a larger 1.08 percentage point 394
395 increase in referrals when controlling for GPA. These are respectively 14 and 20 percent 395
396 increases in the share of referrals received, and suggest participants are able to screen 396
397 higher skilled peers when uniquely referring for cognitive skill. The lower base rate of 397
398 5.4% compared to 7.4% in column (2) suggests less than half of referrals came from 398
399 “unique” referrals. The GPA estimate is five times smaller in magnitude compared 399
400 to column (2), and suggests a smaller weight put on the grades proxy. Nevertheless, 400
401 the comparable magnitudes of GPA and cognitive skill point estimates still suggest 401
402 participants refer peers with higher grades much more often than the correlation between 402
403 the two supported by the data ($\rho = 0.085$). There is heterogeneity in skill identification 403
404 ability even for those who uniquely refer for cognitive skill. 404

405 For participants who receive unique social skill referrals, in column (6), 95% confi- 405
406 dence intervals rule out that a one standard deviation increase in social skill test score 406
407 or GPA result in more than a 0.8 percentage point difference in the share of referrals 407
408 received. These results further support our previous finding that peers cannot screen 408
409 social skills in our sample, and attempts to screen social skills without the GPA proxy 409
410 also seem to fail. 410

411 **Result 3** *The majority of participants cannot screen for skills, refer instead using the 411*
412 *GPA proxy, and nominate the same individuals in common for both skills.* 412

413 **Result 4** *Those who refer uniquely for cognitive skill can identify the skill test score, and 413*
414 *drive the entirety of the results in terms of peer skill identification. Still, they confound 414*
415 *cognitive skill with academic performance, and put similar weights on the two. Those 415*
416 *who refer uniquely for social skill can neither screen social skill or use the GPA proxy.* 416

Table 6: Share of “common” versus “unique” referrals received conditional on skill test score and academic performance

	Common		Unique Cognitive		Unique Social	
	(1)	(2)	(3)	(4)	(5)	(6)
GPA	3.172*** (0.464)	3.670*** (0.501)	0.801** (0.391)	0.752* (0.401)	0.260 (0.334)	0.108 (0.360)
Cognitive score	-0.042 (0.416)	0.139 (0.388)	1.006*** (0.270)	1.084*** (0.281)		
Social score	-0.353 (0.304)	-0.553* (0.272)			0.086 (0.381)	-0.011 (0.357)
Dep. var. mean	7.407	7.382	5.400	5.401	5.485	5.493
Classroom FE	No	Yes	No	Yes	No	Yes
R ²	0.093	0.194	0.028	0.130	0.001	0.090
Observations	665	665	665	665	665	665

Note: Classroom-level clustered standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable in columns (1)-(2) is the percentage share of “common” referrals received from the same referrer, in columns (3)-(4) “unique” referral share for cognitive skill, and in columns (5)-(6) for social skill. Independent variables are the respective standardized test scores for skills and GPA. Sample restricted to 665 individuals for whom we have complete administrative and experimental data.

5.4 Social class bias across common and unique referral types

In this section, we analyze referrals from the perspective of social class while accounting for the referral types described in this part. Based on the referral types from the previous section, we document the existence of a social class bias in referrals when controlling for skill test scores and academic performance at **Baseline**. Our dependent variables are the percentage shares of referrals received at **Baseline** as defined in Equation 6, and we include a social class dummy for the participant receiving the referrals. We estimate for

our three dependent variables:

$$y_i^s = \alpha^s + \beta_1^s GPA_i + \beta_2^s Score_i^s + \beta_3^s SES_i + \epsilon_i^s \quad (7)$$

Table 7 summarizes our findings. Our preferred specification includes classroom fixed effects. The comparison of interest is the SES estimates for the three referral strategies. In column (2), controlling for skill test scores and GPA, the point estimate for low-SES is not statistically significant. Skill score and GPA estimates are robust to the inclusion of this variable and remain close to those in Table 6. However, the marginally significant negative coefficient for social score (-0.524) suggests a slight bias against individuals with higher social skills in common referrals.

For participants who receive unique cognitive skill referrals in column (4), we find that being low-SES causes a 1.8 percentage point decrease in the share of referrals when controlling for cognitive skill and GPA. This is a substantial 28 percent difference in the share of referrals received, confirming participants are biased against low-SES peers when uniquely referring for cognitive skill. Skill test scores and GPA estimates are robust to the inclusion of this variable, but interestingly, the cognitive skill coefficient actually increases from 0.973 compared to Table 6 with the inclusion of the SES dummy. This suggests that, after controlling for socioeconomic status, cognitive skills become even more salient in the referral decision. This is consistent with the data, where low-SES students underperform in cognitive skill tests (see Appendix Figure A.4a). GPA and low-SES are not confounders, and there are no significant differences across social classes in terms of GPA (see Appendix Figure A.3).

For participants who receive unique social skill referrals, in column (6), the point estimate for low-SES is not statistically significant. GPA and social skill estimates remain similar to those in Table 6, although the coefficient on GPA is reduced to near-zero (0.111), suggesting that once we control for both classroom fixed effects and SES, GPA becomes less predictive of social skill referrals. The finding that low-SES students underperform across skill dimensions is consistent with earlier research (Falk, Kosse, Pinger, Schildberg-Hörisch, & Deckers, 2021), though our results show that this bias

manifests most strongly in unique cognitive skill referrals.

Result 5 *We document a sizeable low-SES bias for unique cognitive skill referrals when controlling for cognitive skill test score and academic performance of peers.*

Table 7: Share of “common” versus “unique” referrals received conditional on skill test score, academic performance, and social class

	Common		Unique Cognitive		Unique Social	
	(1)	(2)	(3)	(4)	(5)	(6)
GPA	3.170*** (0.462)	3.663*** (0.499)	0.797** (0.386)	0.766* (0.388)	0.260 (0.334)	0.111 (0.360)
Cognitive score	0.000 (0.411)	0.167 (0.382)	0.869*** (0.261)	0.973*** (0.274)		
Social score	-0.306 (0.315)	-0.524* (0.286)			0.047 (0.372)	-0.027 (0.354)
Low-SES	0.799 (0.939)	0.568 (0.934)	-2.017*** (0.711)	-1.814** (0.713)	-0.549 (0.610)	-0.260 (0.593)
Dep. var. mean	6.948	7.056	6.558	6.442	5.800	5.642
Classroom FE	No	Yes	No	Yes	No	Yes
R ²	0.094	0.194	0.044	0.142	0.002	0.090
Observations	665	665	665	665	665	665

Note: Classroom-level clustered standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable in columns (1)-(2) is the percentage share of “common” referrals received from the same referrer, in columns (3)-(4) “unique” referral share for cognitive skill, and in columns (5)-(6) for social skill. Independent variables are the respective standardized test scores for skills, GPA, and a dummy for low socioeconomic status. Sample restricted to 665 individuals for whom we have complete administrative and experimental data.

5.5 Social class bias and the Quota treatment

In the following empirical specification, we document whether there is a social class bias when referrals are grouped, and whether the **Quota** treatment causes referral shares of low-SES participants to change when controlling for skills and academic performance. We hypothesize that the **Quota** treatment should increase referrals to low-SES peers because of the additional incentive to refer low-SES. The dependent variable is the percentage share of referrals received as defined for the **Baseline** treatment in Equation 1, now extended to the referrals from the **Quota** treatment. It is trivial to see y_{ic}^s can also be calculated for the **Quota** treatment as participants in every classroom are randomized into either treatment. Now, every participant is observed twice in the data, and we add a treatment dummy to indicate whether the referrals came from participants in the **Baseline** or the **Quota** treatment. We also add a social class dummy for the participant receiving the referrals to our specification. We estimate:

$$y_i^s = \alpha^s + \beta_1^s Quota_i + \beta_2^s SES_i + \beta_3^s (Quota_i \times SES_i) + \beta_4^s Score_i^s + \beta_5^s GPA_i + \epsilon_i^s \quad (8)$$

Table 8 illustrates our findings. Our preferred specification includes classroom fixed effects. Our comparison of interest is the effect of the **Quota** treatment on low-SES peers. In column (2) for cognitive skill, we find that being low-SES decreases the share of referrals received by about 1.3 percentage points when controlling for the skill test score and academic performance in the **Baseline** condition. This goes down to about 1.4 percentage points in the **Quota** condition. These differences are not statistically significant, but their sign suggests a relatively large bias against low-SES classmates: A one standard deviation increase in cognitive skill test score has a similar magnitude (0.8 percentage points). This finding is coherent with the bias against low-SES peers found in the previous section, but suggests it is not large enough to carry over to all cognitive skill referrals considered together.

In column (4) for social skill, we find that being low-SES has no statistically significant effect on the share of referrals received when controlling for the skill test score and academic performance. The sign for the point estimate suggests low-SES get slightly

481 more referrals in the **Baseline** condition, and the **Quota** reduces this share. The lack of 481
482 statistical significance in the SES main effect and interaction term suggests considerable 482
483 noise in both estimates. 483

484 **Result 6** *We find no evidence for a general low-SES bias across skills when referrals 484*
485 *are aggregated, when controlling for peer skill test score and academic performance.* 485

Table 8: Share of referrals received by treatment, controlling for skill test score, academic performance, and social class

	Cognitive		Social	
	(1)	(2)	(3)	(4)
Quota	-0.073 (0.755)	-0.073 (0.755)	0.299 (0.716)	0.299 (0.716)
Low-SES	-1.230 (1.079)	-1.276 (1.014)	0.364 (1.282)	0.324 (1.361)
Quota $\times$ Low-SES	-0.167 (1.117)	-0.167 (1.117)	-0.835 (1.181)	-0.835 (1.181)
Score	0.594 (0.448)	0.811* (0.424)	0.201 (0.426)	-0.006 (0.458)
GPA	3.184*** (0.517)	3.522*** (0.552)	2.819*** (0.493)	3.174*** (0.621)
Dep. var. mean	13.551	13.558	12.706	12.714
Classroom FE	No	Yes	No	Yes
R ²	0.060	0.158	0.044	0.134
Observations	1,330	1,330	1,330	1,330

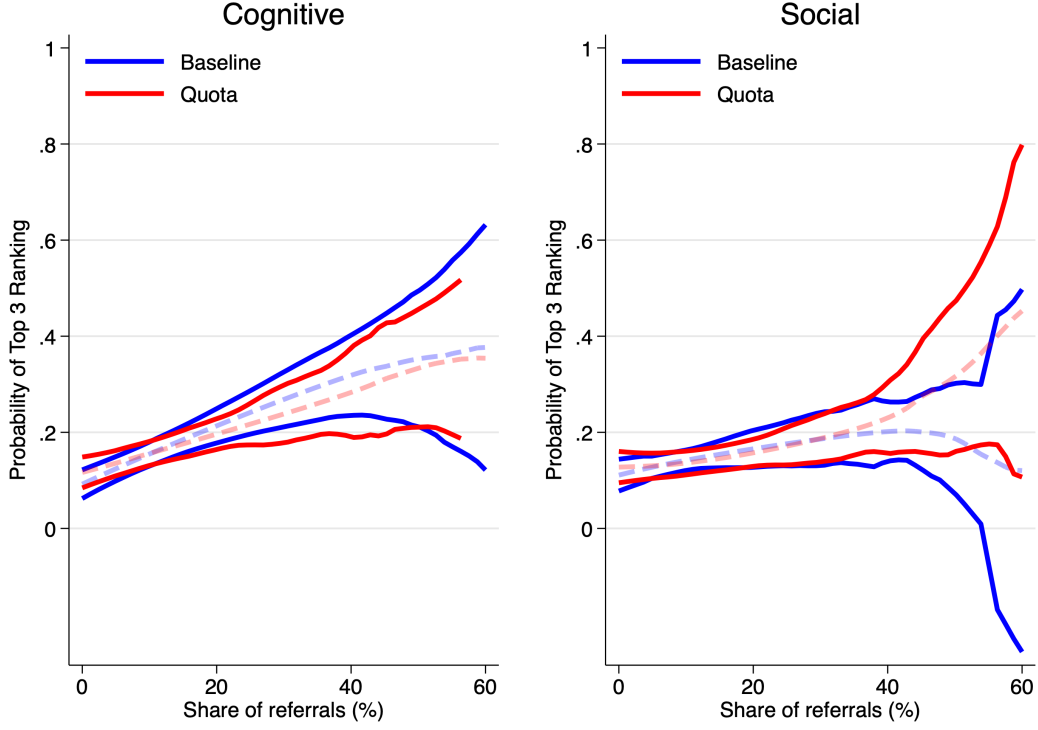
Note: Standard errors in parentheses are clustered at both classroom and individual level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The dependent variables are the percentage share of referrals received for cognitive and social skills. Quota is a dummy for the referrals received from classmates in the **Quota** treatment. Low-SES is a dummy for participant's socioeconomic status. Remaining independent variables are the respective standardized test scores for skills and GPA. Sample includes 1,330 observations with complete administrative and experimental data.

5.6 Quota treatment and the productivity of the referrals

The equity-efficiency tradeoff is a valid concern for the **Quota** treatment, as any intervention that changes the referral decisions in terms of SES composition should not reduce

the productivity of the pool of referred candidates. To address this concern, in Figure 6, we plot the share of referrals received across the two conditions and the probability of being among the Top 3 in the classroom for either skill. We find first that the slopes of the distributions are always positive for the **Quota** treatment, indicating a positive relationship with the share of referrals received. Second, a two-sample Kolmogorov-Smirnov test reveals no statistically significant differences in the distribution of referrals between the two conditions for both cognitive and social referrals. These suggest that the **Quota** treatment does not reduce the productivity of the pool of referred candidates.

Figure 6: Referral shares and the probability of being in the Top 3



Note: The left panel shows how Baseline and Quota referral shares vary with the probability of being in the Top 3 of the classroom for cognitive skill scores, while the right panel shows the same figure for social skill scores. Solid lines indicate the 95% confidence intervals, with dashed lines representing means. The output is truncated at 60 percent of referral share to ensure meaningful confidence intervals. Two-sample Kolmogorov-Smirnov tests reveal no statistically significant differences in the distribution of referrals between Baseline and Quota conditions for both cognitive referrals ($D = 0.0351$, $p = 0.710$) and social referrals ($D = 0.0439$, $p = 0.427$).

5.7 Effects of the Quota treatment across referral types

Effects of the social class bias gets diluted across common and unique referral types. A large proportion of participants -“common” referrers- who struggle with skill identification and screen for skills using the academic performance proxy. But there are no SES

differences for GPA in our sample. When referrals are made with academic performance in mind, it seems reasonable not to observe a negative bias against low-SES. Then what about skills, knowing that high-SES score higher in both measures?

We observe a bias in undersampling from equally well performing low-SES only for “unique” cognitive skill referrals, where referrers screen better compared to “unique” social skill referrals. We expect the **Quota** treatment be effective in increasing referrals for low-SES only in a scenario where the skill can be screened, and turn toward our classification of different referral types to test this hypothesis. To get clearer estimates for the effects of the **Quota** treatment on low-SES referrals, we re-estimate the shares of “common” and “unique” referrals. Following the same logic in the section before, we observe every participant twice in each specification, and add a treatment dummy to indicate whether the referrals came from referrers in the **Baseline** or the **Quota** treatment. We keep the social class dummy and regress Equation 8 for the three dependent variables.

Table 9 illustrates our findings. Our preferred specification includes classroom fixed effects. The comparison of interest is the SES of the participant receiving the referrals and the effect of the **Quota** treatment across “common” and “unique” referral types. In column (2), for participants who refer the same peers in common using the academic performance proxy, we find no statistically significant effect of participant SES or the **Quota** on the referrals share when controlling for skill test scores and academic performance. The signs on the point estimates suggest that directionally low-SES get more referrals in the **Baseline** treatment and that the **Quota** reduces this share.

For participants who make unique cognitive skill referrals, in column (4), we find that being low-SES in the **Baseline** treatment reduces the percentage share of referrals received by 1.9 percentage points when controlling for the skill test score and academic performance. This is a very large effect size which translates to a decrease in referral share by 29 percent on a base rate of 6.5%, and is similar to the one found in Table 7. In turn, the **Quota** treatment increases referrals to low-SES by 1.42 percentage points when controlling for the skill test score and academic performance. This is also a large

530 effect size that results in an increase in low-SES referral share by 22 percent. 530

531 For participants who make unique social skill referrals, in column (6), we find no 531
532 statistically significant effect of participant SES or the **Quota** on the referrals share when 532
533 controlling for the skill test score and academic performance. These are in accordance 533
534 with our previous findings that social skills cannot be identified and provide no reason 534
535 to warrant social class bias for this skill domain. 535

536 **Result 7** *There is a bias against low-SES peers only for the skill that is well-identified* 536
537 *by peers, and in which low-SES underperform. We find no evidence of a bias when* 537
538 *referrals are made based on academic performance where both social classes perform* 538
539 *equally well.* 539

540 **Result 8** *The bias in unique cognitive skill referrals is partially alleviated by the **Quota*** 540
541 *treatment. Because there is remarkable heterogeneity in the ability to detect SES for* 541
542 *both social classes (see Appendix Figure A.5), we find this increase in low-SES referrals* 542
543 *satisfying for our sample.* 543

Table 9: Share of “common” and “unique” referrals received by treatment, controlling for skill test score, academic performance, and social class

	Common		Unique Cognitive		Unique Social	
	(1)	(2)	(3)	(4)	(5)	(6)
Quota	0.436 (0.817)	0.436 (0.817)	-0.509 (0.598)	-0.509 (0.598)	-0.136 (0.523)	-0.136 (0.523)
Low-SES	0.857 (0.920)	0.598 (0.897)	-2.074*** (0.722)	-1.891** (0.710)	-0.510 (0.613)	-0.256 (0.594)
Quota $\times$ Low-SES	-1.584 (1.159)	-1.584 (1.159)	1.417** (0.656)	1.417** (0.656)	0.750 (0.717)	0.750 (0.717)
Cognitive score	-0.079 (0.374)	0.095 (0.346)	0.658*** (0.201)	0.739*** (0.210)		
Social score	0.062 (0.283)	-0.091 (0.236)			0.158 (0.312)	0.061 (0.269)
GPA	2.322*** (0.330)	2.727*** (0.366)	0.858*** (0.312)	0.804** (0.340)	0.502* (0.278)	0.439 (0.292)
Dep. var. mean	6.952	7.080	6.591	6.488	5.765	5.623
Classroom FE	No	Yes	No	Yes	No	Yes
R ²	0.052	0.139	0.028	0.099	0.005	0.071
Observations	1,330	1,330	1,330	1,330	1,330	1,330

Note: Standard errors in parentheses are clustered at both classroom and individual level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable in columns (1)-(2) is the percentage share of “common” referrals received from the same referrer, in columns (3)-(4) “unique” referral share for cognitive skill, and in columns (5)-(6) for social skill. Quota is a dummy for the referrals received from classmates in the **Quota** treatment. Low-SES is a dummy for participant’s socioeconomic status. Remaining independent variables are the respective standardized test scores for skills and GPA. Sample includes 1,330 observations with complete administrative and experimental data.

6 Conclusion

In this paper, we study how accurately individuals assess productivity of their peers across different skill dimensions and whether these assessments systematically disadvantage low-SES individuals in a diverse university setting. Through a lab-in-the-field experiment that isolates screening ability, we find that the accuracy of peer productivity assessments varies significantly across skill types, with implications for referral-based screening.

Our findings reveal that peers can effectively identify cognitive skills but struggle to assess social skills in their classmates. This differential screening ability appears to stem from the inherent challenges in evaluating interpersonal capabilities compared to cognitive abilities. When faced with uncertainty in skill assessment, peers often rely on observable proxies like academic performance which may be misleading. This suggests that the effectiveness of peer assessments depends crucially on how discernible the target skill is, rather than indicating a fundamental limitation of referrals as a screening mechanism.

These results complement the broader literature showing referrals' effectiveness in worker screening by highlighting how skill visibility affects assessment accuracy. While previous work demonstrates that referrals successfully identify productive workers overall (Pallais & Sands, 2016), our findings suggest their effectiveness may vary across different dimensions of human capital. This variation is particularly relevant given the growing importance of social skills in labor markets as found in other research (Deming, 2017). Our evidence also supports earlier evidence that accurate assessment of social skills remains challenging (Bassi & Nansamba, 2022), suggesting the need for either longer periods of interaction to discern these skills or development of alternative assessment methods that can better capture interpersonal capabilities in referral settings.

Looking forward, our findings suggest several implications for improving screening mechanisms in similar settings. First, institutions that implement referral programs may need to develop complementary tools for evaluating less visible skills like inter-

572 personal capabilities, perhaps in the likes of the social skill certificates in [Bassi and](#) 572
573 [Nansamba \(2022\)](#). Second, our results on social class bias - finding it only in unique cog- 573
574 nitive skill referrals and its mitigation through quota incentives without compromising 574
575 productivity- indicate that targeted interventions can effectively address specific biases 575
576 without compromising the overall screening process. Future research could investigate 576
577 how to optimize referral programs to leverage their strengths in identifying easier to 577
578 discern skills while developing better methods for assessing harder to observe skills, par- 578
579 ticularly given the increasing importance of these interpersonal capabilities in modern 579
580 labor markets. 580

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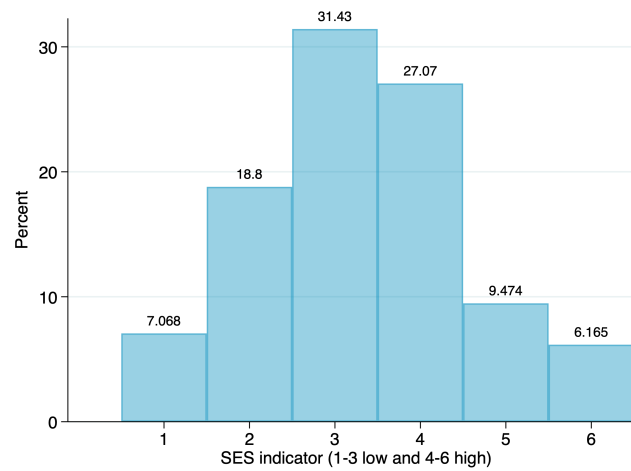
701 A Additional Figures and Tables

701

702 A.1 Additional Figures

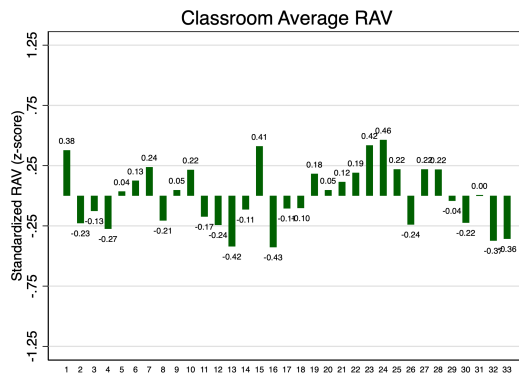
702

Figure A.1: Stratum distribution of the sample

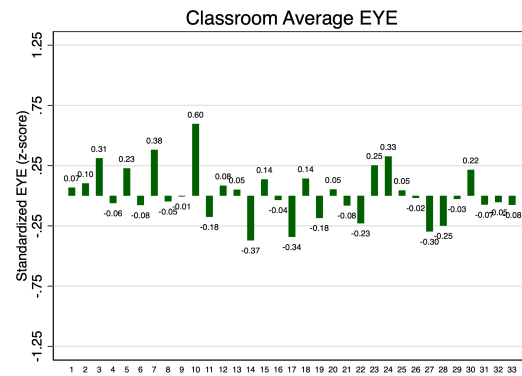


Note: This figure shows the distribution of strata in the sample of students that participated in the study.

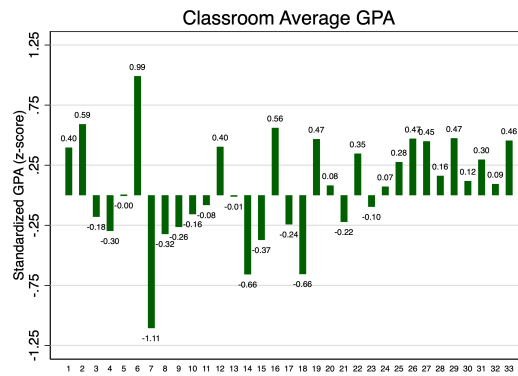
(a) Cognitive score across classrooms



(b) Social score across classrooms

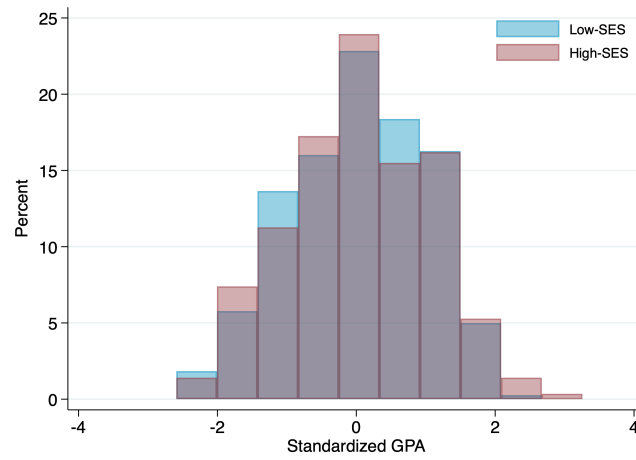


(c) GPA across classrooms



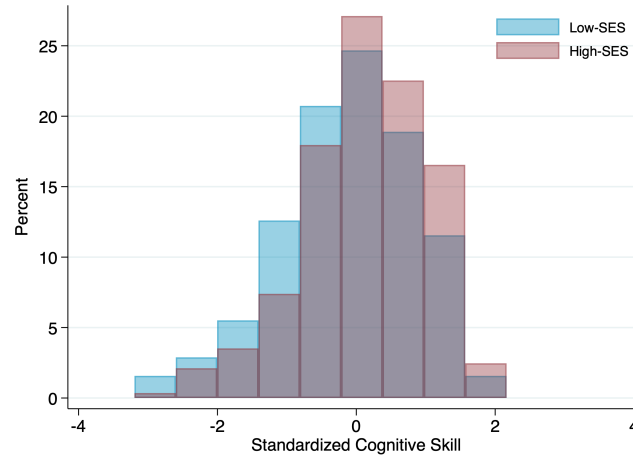
Note: These figures show the respective distribution of standardized scores for cognitive skill, social skill, and GPA across sampled classrooms.

Figure A.3: GPA by SES

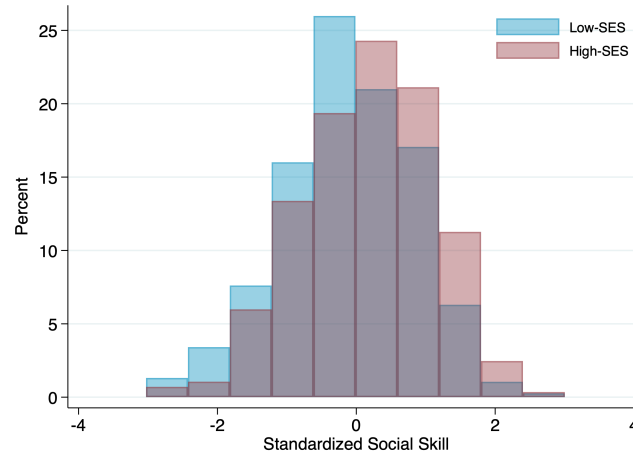


Note: This figure shows the distribution of GPA across SES. There are no significant differences in the mean standardized GPA scores between high-SES and low-SES participants (t test $p = 0.695$).

(a) Cognitive score by SES

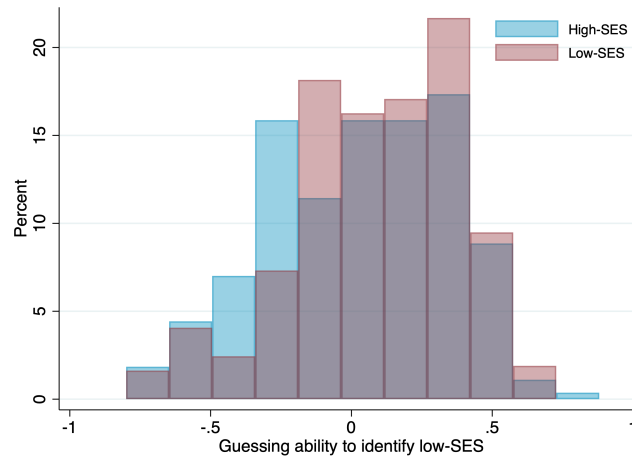


(b) Social score by SES



Note: These figures show the respective distribution of cognitive and social skills across SES. High social class outperform Low-SES in both skills (t tests have p values < 0.001). We can visually verify that larger share of high-SES in quantiles above median for both skills.

Figure A.5: Distribution of guessing ability across SES



Note: This figure shows the distribution of the guessing ability across SES. We calculate the guessing ability as the share of succesful low-SES guesses minus the expected probability of randomly drawing low-SES in class c . A score of 0 indicates an accuracy as good as random draws, below 0 drawing worse than chance, and above 0 better than chance. There are significant differences in the mean guessing ability between high-SES ($M = 0.022$, $SD = 0.325$, $n = 271$) and low-SES participants ($M = 0.093$, $SD = 0.302$, $n = 369$), $t(638) = -2.85$, $p = 0.005$, $d = 0.226$. Low-SES participants have higher guessing ability compared to their high-SES counterparts, with a mean difference of 7 percentage points.

Table A.1: Comparison of Variables Between the Sample and Missing Students

	Sample	Missing	<i>p</i>
Referral share (both skills)	0.127	0.043	0.000
GPA (standardized)	0.044	-0.273	0.001
Entry Exam (standardized)	0.028	-0.168	0.046
# Semesters at UNAB	3.171	3.188	0.884
Age	19.182	20.287	0.001
Female	49.8%	48.5%	0.788
Ethnic Minority	2.1%	4.4%	0.114
Rural Community	28.8%	31.6%	0.501
Has Scholarship	0.8%	0.7%	0.899

Note: Values for female, ethnic minority, rural community, and scholarship represent percentage proportions. All other variables represent means. *p*-values for gender, ethnic, rural, and scholarship are from two-sample tests of proportions. For all other variables, *p*-values are from two-sample t-tests with equal variances. All tests compare the sample and missing students. All reported *p*-values are two-tailed.

Table A.2: Correlation between GPA, entry exam, and skill test scores

	GPA	Cognitive score	Social score	Entry Exam
GPA	1.000			
Cognitive score	0.083	1.000		
Social score	0.091	0.266	1.000	
Entry Exam	0.229	0.403	0.267	1.000

Note: Pairwise correlation between GPA, entry exam, and skill test scores. Sample is restricted to 655 participants with complete administrative and experimental data.

Table A.3: Between-Classroom Variation in Academic Programs

Statistic	Most common program share
Mean	0.424
Standard Deviation	0.216
10th percentile	0.174
25th percentile	0.292
Median	0.345
75th percentile	0.533
90th percentile	0.696
# classrooms with share 1	3
Most diverse classroom	0.154
# classrooms	35

Note: Table shows the distribution of academic programs across classrooms, measured by the share of students from the most common program in each classroom. Three classrooms are completely homogeneous (share = 1). In the median classroom, the most common program accounts for 34.5% of students. The most diverse classroom has only 15.4% of students in the same program. Data based on 849 students across 35 classrooms.

Table A.4: Characteristics of self-referrers

	No self-referral	Any self-referral	Δ	p
GPA	0.132 (1.003)	-0.120 (0.966)	0.252	0.002
Cognitive score	0.087 (0.988)	-0.118 (1.023)	0.205	0.013
Social score	0.034 (1.003)	-0.038 (0.959)	0.072	0.374
Low-SES	0.605 (0.490)	0.511 (0.501)	0.094	0.021
N	440	225	665	
Share (%)	66.2	33.8	100	

Note: Table compares standardized scores between participants who self-referred at least once ($N = 225$) and those who did not ($N = 440$). Positive differences indicate higher scores for those who never self-referred. p -values from two-sided t -tests (GPA, Cognitive Skill, Social Skill) and proportion test (Low-SES). The results suggest self-referrers have significantly lower cognitive skills and GPA, and are more likely to be low-SES. Standard deviations in parentheses, samples restricted to participants with complete administrative and experimental data.

Table A.5: Characteristics of participants who make overlapping referrals

	Unique referrals	Common referrals	Δ	p
GPA	0.057 (0.983)	0.045 (1.009)	0.012	0.903
Cognitive score	0.110 (1.005)	0.024 (0.979)	0.086	0.371
Social score	-0.014 (0.938)	0.033 (0.981)	-0.047	0.621
Low-SES	0.530 (0.501)	0.597 (0.491)	-0.067	0.164
N	132	512	644	
Share (%)	20.5	79.5	100	

Note: Table compares characteristics between participants who made at least one overlapping referral ($N = 512$) to those who did not ($N = 132$, 20.5%). Overlapping referrals indicate cases where a participant referred the same classmate once for cognitive or social skills. Positive differences indicate higher scores for those who made no overlapping referrals. The results suggest minimal differences across all variables. p -values from two-sided t-tests (GPA, Cognitive Skill, Social Skill) and proportion test (Low-SES). Standard deviations in parentheses, sample restricted to participants with complete administrative and experimental data.

Table A.6: Characteristics of Top Performers and Referrals

	Cognitive		Social		Both
	Top 3	Referrals	Top 3	Referrals	Top 3
Cognitive score	1.223 (0.419)	0.112 (1.009)	0.383 (0.922)	0.058 (1.015)	1.201 (0.458)
Social score	0.357 (0.923)	0.086 (0.996)	1.340 (0.395)	0.042 (1.009)	1.391 (0.453)
GPA	0.277 (0.990)	0.251 (1.021)	0.264 (1.046)	0.212 (1.004)	0.551 (0.897)
Low-SES	0.457 (0.500)	0.532 (0.499)	0.456 (0.500)	0.555 (0.497)	0.500 (0.507)
N	129	1,759	114	1,775	36
Share (%)	20.0	100	17.7	100	5.6

Note: Table shows characteristics of students ranked in the top 3 of their classroom and average characteristics of referred students, by skill. Standard deviations in parentheses. Sample restricted to participants with complete administrative and experimental data. All continuous variables are standardized.

704 B Experiment 704

705 *We include the English version of the instructions used in Qualtrics. Participants saw* 705
706 *the Spanish version. Horizontal lines indicate page breaks, and clarifying comments are* 706
707 *inside brackets.* 707

708 _____ 708

709 709

710 Please enter the password: 710

711 [classroom-specific password sent to each participant the day before data collection] 711

712 _____ 712

713 713

714 **Welcome** 714

715 715

716 Welcome to this study organized by the Social Bee Lab. You have been invited to 716

717 participate in a survey where you can make a series of decisions. The study takes ap- 717

718 proximately 20 minutes to complete. During the study, you should not communicate 718

719 with any other students. If you have any questions at any time, please raise your hand. 719

720 One of the assistants will help you privately. 720

721 721

722 In this study, you can win bonus money depending on your choices. In total, we will 722

723 draw [classroom-specific number equal to 40% of class size] bonuses of 100.000 pesos 723

724 among the participants of this classroom. It is also possible for the same person to win 724

725 more than one voucher. The following screens will detail how the bonus draw will be 725

726 conducted. The UNAB finance office will make the payment of the vouchers through 726

727 Nequi. 727

728 728

729 All your decisions in this survey will be anonymized. Therefore, the answers you provide 729

730 will not affect your grades in this class or your records at the university. We will use your 730

731 personal information to determine the bonus allocation, but after that, we will remove 731

any data that identifies you.

This survey has several parts. Each of these parts has specific instructions. Please read the instructions for each part carefully because they describe how you can earn bonuses. This study has been approved by the [omitted for anonymous review] on the condition that all the information we provide is true and all the bonuses we offer are real. On the next screen, we present you with an informed consent form that you must accept to participate in this study.

Informed Consent

You have been invited to participate in a study to learn more about how people make decisions in common scenarios.

This study is conducted by [omitted for anonymous review] and the Social Bee Lab at UNAB. The purpose of this study is to broaden our understanding of how people make decisions.

Participation in this study is voluntary. You may opt-out at any time. No known risks are associated with your participation in this project beyond those of everyday life. Apart from the monetary bonuses that will be drawn, participation has no direct benefits.

The Social Bee Lab is in charge of data collection. Your answers in this study are anonymous and will not be shared with anyone. In addition to your answers, UNAB will provide the Social Bee Lab with administrative records of your courses and your university entrance exam score. Your records, decisions, and your identity will be kept strictly confidential. Data about you collected within the scope of the study are used for scientific purposes only and are treated as strictly confidential. The Social Bee Lab will

761 anonymize your data, and the researcher will analyze it without knowing your identity. 761
762 All data generated will be stored on the researcher's computer. You have the right to 762
763 access your personal data and request its deletion. You can exercise this right by con- 763
764 tacting the researcher. 764
765 765
766 If something is unclear or you have any questions, you can contact [omitted for anony- 766
767 mous review]. 767
768 768
769 If you have questions about your rights as a participant, you can contact [omitted for 769
770 anonymous review]. 770
771 771
772 By continuing to the next screen, you agree to participate in this study. 772
773 773
774

 774
775 775
776 Before you start, please answer these four questions. 776
777 777
778 What is your gender? 778
779 779
780 [Male, Female] 780
781 781
782 What is the socio-economic stratum to which your family belongs? 782
783 783
784 [Stratum 1 to Stratum 6] 784
785 785
786 What is your father's highest acquired level of education? 786
787 787
788 [Primary school, High school, Technical school, Undergraduate, Graduate, Postgradu- 788
789 ate, Not applicable.] 789

790 790

791 What is your mother's highest acquired level of education? 791

792 792

793 [Primary school, High school, Technical school, Undergraduate, Graduate, Postgradu- 793

794 ate, Not applicable.] 794

795 795

796 _____ 796

797 797

798 **Part 1** 798

799 799

800 You will now participate in two quizzes, each lasting five minutes. Please try to answer 800

801 them to the best of your ability. 801

802 802

803 We will allocate up to [classroom-specific number equal to 20% of class size] bonuses of 803

804 100.000 pesos in this first part. The steps to allocate the bonuses for Part 1 are explained 804

805 below. 805

806 806

807 _____ 807

808 808

809 [classroom-specific illustrations explaining the incentive structure] 809

810 810

811 _____ 811

812 812

813 [random assignment to either cognitive or social skills test] 813

814 814

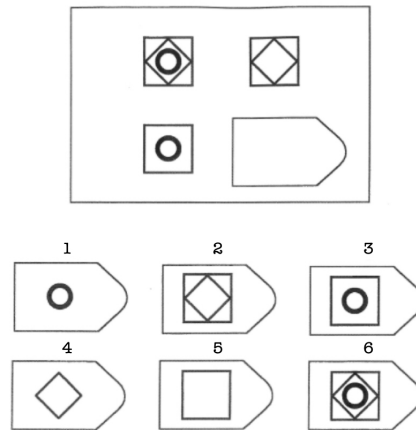
815 **Test - Cognitive Skill** 815

816 816

817 In this test, you will see a series of images. Below is an example of the images you will 817

818 solve. At the top of each image, there is a pattern with a piece that has been removed. 818

819 Your task is to choose which of the six pieces completes the pattern correctly. For each 819
 820 image, there is only one correct piece. Look at the following example: 820
 821 821



822 First, notice a square in the upper left, the upper right, and the lower left. Also, notice 822
 823 that the circle is eliminated when one moves from the upper left to the upper right. 823
 824 Finally, the rhombus is eliminated when moving from the upper left to the lower left. 824
 825 Therefore, the correct piece should eliminate the circle and the rhombus, leaving only a 825
 826 square. So, the correct answer is piece 5. 826

827 827
 828 To give your answer to each image, you must choose the correct option and then continue 828
 829 to the next screen. After giving your answer you cannot go back. 829

830 830
 831 You will have 5 minutes to complete the test, which consists of 18 images to solve. The 831
 832 percentage of correct answers will determine your chances of winning one of the 100.000 832
 833 pesos bonuses if you are chosen for the drawing. 833

834 834

835 _____ 835

836 836

837 Are you ready? 837

838		838
839	Your 5 minutes will start as soon as you move to the next screen.	839
840		840
841		841
842		842
843	Problem 1	843
844		844
845	[screenshot of Raven's matrix]	845
846		846
847	[After participants submit an answer, a new matrix appears on the screen. The se-	847
848	quence of matrices is the same for all participants. Participants cannot return to a	848
849	previous screen. Participants do not have to provide answers for all 18 matrices.]	849
850		850
851		851
852		852
853	You have finished the test. You can proceed to the next screen.	853
854		854
855		855
856		856
857	How did you do on the test?	857
858		858
859	If we randomly choose 10 participants from this classroom, how many people do you	859
860	think solved fewer correct problems than you?	860
861		861
862	[Slider from 0 to 10]	862
863		863
864		864
865		865
866	Test - Emotions	866

867

867

868 In this test, you will see a series of photographs. Below is an example of the pictures you 868
869 will see. In each picture, you will see the eyes of a person. Below the picture, you will 869
870 see four possible emotions that this person is feeling. Your task is to choose which of 870
871 the four emotions correctly describes what the person is feeling. For each picture, there 871
872 is only one emotion. Look at the following example: 872

873

873



874 [Happy, Disappointed, Shocked, Worried] 874

875

875

876 In this case, the correct answer is: Shocked. 876

877

877

878 To give your answer to each picture, you must choose the correct option and then con- 878
879 tinue to the next screen. After giving your answer you will not be able to go back. 879

880

880

881 You will have 5 minutes to complete the test, which consists of 36 photographs to solve. 881
882 The percentage of correct answers you get will determine your chances of winning one 882
883 of the 100.000 pesos bonuses if you are chosen for the drawing. 883

884

884

885 _____ 885

886

886

887 Are you ready? 887

888

888

889 Your 5 minutes will start as soon as you move to the next screen. 889

890

890

891 _____ 891

892 892

893 **Photograph 1: Choose the word that best describes the photograph** 893

894 894

895 [photo from Multiracial Reading the Mind in the Eyes Test] 895

896 896

897 [After participants submit an answer, a new photo appears on the screen. The sequence 897

898 of photos is the same for all participants. Participants cannot return to a previous screen. 898

899 Participants do not have to provide answers for all 36 photos.] 899

900 900

901 _____ 901

902 902

903 You have finished the test. You can proceed to the next screen. 903

904 904

905 _____ 905

906 906

907 **How did you do on the test?** 907

908 908

909 If we randomly choose 10 participants from this classroom, how many people do you 909

910 think solved fewer correct photographs than you? 910

911 911

912 [Slider from 0 to 10] 912

913 913

914 _____ 914

915 915

916 **Part 2** 916

917 917

918 At the beginning of this study, all participants took two tests, one on cognitive ability 918

919 and one on emotions. In this part, we will ask you to recommend the people who in 919

920 your opinion will score the best on each test. 920

921 You may recommend 3 people per test, but you may not recommend yourself. 922

923 We will allocate up to [classroom-specific number equal to 40% of class size] bonuses of 924

925 100.000 pesos for Part 2. The steps for allocating bonuses are explained below. 926

927 _____ 928

929 [random assignment to either quota or baseline condition] 930

931 [classroom-specific illustrations explaining the incentive structure depending on assign- 932

933 ment to either baseline or quota conditions] 934

Figure B.1: Illustrations for the two conditions

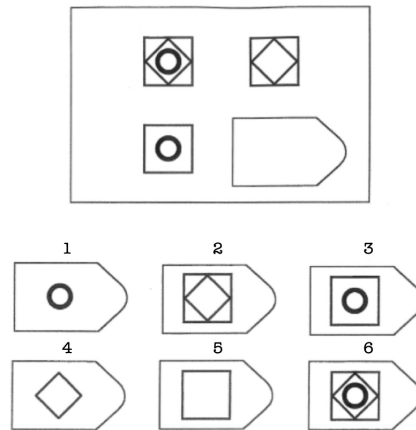


934 _____ 935

936 [random assignment to either cognitive or social skills referral task] 937

938 **Recommendation - Cognitive Skill** 939

940 All participants took a test to identify the missing pattern in each image, as in the ex- 940
941 ample below. This test is used to measure general intelligence. 941
942 942



943 Next, we will present you with a list of the names of all the students in this room. We 943
944 will ask you to recommend the three people you think will score the highest on the 944
945 general intelligence test. 945

946 946
947 If you are chosen by the computer, each of your recommendations in the top 3 increases 947
948 your chances of winning one of the 100.000 pesos bonuses. 948

949 949
950 _____ 950
951 951

952 Select the students in this classroom who you consider to have the highest scores on the 952
953 general intelligence test. (Select 3 students) 953

954 954

955 [Classroom-specific list of all classmate names visible on one screen. Participants have 955
956 to pick 3 classmates to continue. Picking their own name invalidates their choices.] 956

957 957

958 _____ 958

959

959

960 **Recommendation - Emotions**

960

961

961

962 All participants took a test where they had to identify the emotion that best described
963 the expression of each image as in the example below. This test is used to measure social
964 skills.

962

963

964

965

965



966 Next, we will present you with a list of the names of all the students in this room. We
967 will ask you to recommend 3 people you think will score the highest on the social skills
968 test.

966

967

968

969

969

970 If you are chosen by the computer, each of your recommendations in the top 3 increases
971 your chances of winning one of the 100.000 pesos bonuses.

970

971

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972

973 _____

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974

975 Select the students in this classroom who you consider to have the highest scores on the
976 social skills test. (Select 3 students)

975

976

977

977

978 [Classroom-specific list of all the names visible on one screen. Participants have to pick
979 3 classmates to continue. Picking their own name invalidates their choices.]

978

979

980

980

981 _____

981

982

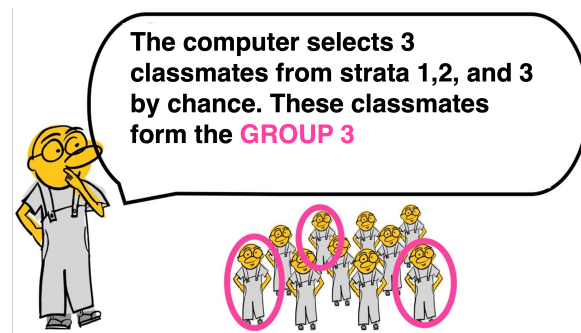
982

983 **Part 3: Recommendation - Random draw**

983

984 In this part, the computer will randomly choose three students who belong to strata 1, 984
 985 2, or 3. We will ask you to nominate three people you think the computer will choose. 985
 986 986
 987 We will allocate up to [classroom-specific number equal to 20% of class size] bonuses of 987
 988 100.000 pesos for Part 3. The steps for allocating the bonuses are explained below. 988
 989 989
 990 _____ 990
 991 991
 992 [classroom-specific illustrations explaining the incentive structure] 992
 993 993

Figure B.2: Illustration for the Guessing Task



994 _____ 994
 995 995
 996 Select the students in this classroom who belong to strata 1, 2, or 3, who you think will 996
 997 be randomly selected by the computer (Select 3 students). 997
 998 998
 999 [Classroom-specific list of all the names visible on one screen. Participants have to pick 999
 1000 3 classmates to continue. Picking their own name invalidates their choices.] 1000
 1001 1001
 1002 _____ 1002
 1003 1003

1004 **Part 4** 1004

1005 Do you want to know your scores on the general intelligence test and the social skills 1005

1006 test? We can analyze the data and give you a report that explains your strengths in 1006

1007 these two areas. Also, what do these strengths mean, and how can you leverage them 1007

1008 for your personal and professional development? 1008

1009 1009

1010 If you want to receive your skills report, we need to contact you again. We also want to 1010

1011 be able to invite you to new studies where you can participate for more bonus money. 1011

1012 Please indicate if you agree to be contacted again. 1012

1013 1013

1014 [I can be contacted for new studies and to send me my report. I can be contacted to 1014

1015 send my report, but not for new studies. No, I do not want to be contacted again.] 1015

1016 _____ 1016

1017 1017

1018 [if participant gives consent to be contacted again] 1018

1019 1019

1020 Please enter your contact email: 1020

1021 1021

1022 [student email] 1022

1023 1023